

Monday
5/10/26. ✓

time per instruction.

Latency (same)

WAR

This is only problematic if the above two instr. are executed out of order BUT RISC-V does not allow it.

add x11 ...

Sub $x/11 \dots$

* WAR and WAW are data hazards but only RAW only comes in RISC $\rightarrow$ which is "Time dependency."

Question: rd $x_{10}, 0(x_{12})$
 sub x_{10}, x_{10}, x_{12}
 add x_{10}, x_{12}, x_{14}

How many cycles with
8 about forwarding.

W/O Forwarding: we get result after WB.

Clock Cycles →	1	2	3	4	5	6	7	8	9	10	11
ld	IF	ID	EX	MEM	WB	✓					
Sub		IF	ID	bubble	□	Ex	MEM	WB	✓		
add			IF	□	□	ID	□	□	Ex	MEM	WB

IF $\leftarrow$ IF of next ms. will be from here ages
koi ho.

Total C.C w/out forwarding = 11

★ Bubbles will be in entire column on other instr. too. as CPU is in 'No operation stage'.

- With forwarding:

In load; we get result after MEM

Clock cycles →	1	2	3	4	5	6	7	8
rd	IF	ID	EX	M	WB	✓		
sub		IF	ID	□	EX	M	WB	✓
add			IF	□	ID	EX	M	WB

Total with forwarding = 8 clock cycles.

NOTE: Dependency (RAW) could be b/w first and fifth instr. as well.

Q2: ld x10, 0(x8)

sub x8, x11, x12

ld x12, 0(x10)

	1	2	3	4	5	6	7	8
Ld	IF	ID	EX	MEM	WB			
Sub		IF	ID	EX	M	WB		
Ld			IF	ID	□	EX	M	WB

Without forwarding = 8-CC

Code Scheduling: To insert the instr. which doesn't have dependencies (Independ ones like sub.) between the two instructions which have dependencies. This gives the time for 2nd dependency instr. to get data result from above.